Python Assignment for Data Science / Data Engineer / Data Analyst

Complete solutions to all 17 questions. Outputs are shown for the small examples used in each program; file-based answers depend on the user's input file.

Q1. Sum and average of a list

Python / SQL code

```
numbers = [10, 20, 30, 40, 50]
total = sum(numbers)
average = total / len(numbers)
print("Sum:", total)
print("Average:", average)
```

Sample output

```
Sum: 150
Average: 30.0
```

Q2. List, tuple and dictionary

A list is ordered and mutable; a tuple is ordered and immutable; a dictionary stores key-value pairs and is mutable.

Python / SQL code

```
items = [10, 20, 30]
point = (10, 20, 30)
student = {"name": "Asha", "age": 21}
print(items[0], point[1], student["name"])
items.append(40)
student["age"] = 22
# point[0] = 99  # TypeError: tuple does not support item assignment
```

Sample output

```
10 20 Asha
List after append: [10, 20, 30, 40]
Dictionary after update: {'name': 'Asha', 'age': 22}
```

Q3. Load CSV into Pandas and display first 5 rows

Replace data.csv with the path to your CSV file.

Python / SQL code

```
import pandas as pd
df = pd.read_csv("data.csv")
print(df.head())
```

Sample output

```
Displays the first five rows of data.csv. Exact rows depend on the supplied CSV file.
```

Q4. Handling missing values

Inspect missingness first; then drop or impute values according to the meaning of each column.

Python / SQL code

```
print(df.isna().sum())
df["Age"] = df["Age"].fillna(df["Age"].median())
df["City"] = df["City"].fillna("Unknown")
df = df.dropna(subset=["Target"])  # if target is required
```

Sample output

```
Missing-value counts are printed for every column.
Age missing values are replaced by the median; City by 'Unknown';
rows with a missing Target are removed.
```

Q5. Count word frequency in a text file

Python / SQL code

```
from collections import Counter
import re

def word_frequency(filename):
```

```
    with open(filename, "r", encoding="utf-8") as file:
        text = file.read().lower()
    words = re.findall(r"\b[\w']+\b", text)
    return dict(Counter(words))

print(word_frequency("sample.txt"))
```

Sample output

```
For sample.txt containing: "Data science is fun. Data is useful."
{'data': 2, 'science': 1, 'is': 2, 'fun': 1, 'useful': 1}
```

Q6. Fibonacci using recursion

Python / SQL code

```
def fibonacci(n):
    if n < 0:
        raise ValueError("n must be non-negative")
    if n <= 1:
        return n
    return fibonacci(n - 1) + fibonacci(n - 2)

print([fibonacci(i) for i in range(8)])
```

Sample output

```
[0, 1, 1, 2, 3, 5, 8, 13]
```

Q7. Check even or odd

Python / SQL code

```
n = 17
if n % 2 == 0:
    print("Even")
else:
    print("Odd")
```

Sample output

```
Odd
```

Q8. Swap without a third variable

Python / SQL code

```
a, b = 5, 9
a, b = b, a
print("a =", a, "b =", b)
```

Sample output

```
a = 9 b = 5
```

Q9. Factorial

Python / SQL code

```
def factorial(n):
    if n < 0:
        raise ValueError("Factorial is undefined for negative integers")
    result = 1
    for value in range(2, n + 1):
        result *= value
    return result

print(factorial(5))
```

Sample output

```
120
```

Q10. Palindrome string

Python / SQL code

```
def is_palindrome(value):
    cleaned = ''.join(ch.lower() for ch in value if ch.isalnum())
    return cleaned == cleaned[::-1]

print(is_palindrome("Madam"))
```

Sample output

```
True
```

Q11. Largest number in a list

Python / SQL code

```
numbers = [12, 45, 7, 89, 23]
print(max(numbers))
```

Sample output

```
89
```

Q12. Remove duplicates

Python / SQL code

```
values = [3, 1, 3, 2, 1, 4]
unique_values = list(dict.fromkeys(values))  # preserves order
print(unique_values)
```

Sample output

```
[3, 1, 2, 4]
```

Q13. Count occurrences of each element

Python / SQL code

```
from collections import Counter
values = [1, 2, 2, 3, 1, 1]
print(dict(Counter(values)))
```

Sample output

```
{1: 3, 2: 2, 3: 1}
```

Q14. Count vowels

Python / SQL code

```
text = "Data Science"
count = sum(ch.lower() in "aeiou" for ch in text)
print(count)
```

Sample output

```
5
```

Q15. Reverse each word in a sentence

Python / SQL code

```
sentence = "Python is easy"
result = " ".join(word[::-1] for word in sentence.split())
print(result)
```

Sample output

```
nohtyP si ysae
```

Q16. Merge two dictionaries

Python / SQL code

```
first = {"a": 10, "b": 20}
second = {"b": 5, "c": 30}
merged = first.copy()
for key, value in second.items():
    merged[key] = merged.get(key, 0) + value
print(merged)
```

Sample output

```
{'a': 10, 'b': 25, 'c': 30}
```

Q17. Read a file and count words

Python / SQL code

```
with open("sample.txt", "r", encoding="utf-8") as file:
    text = file.read()
word_count = len(text.split())
```

```
print("Word count:", word_count)
```

Sample output

```
For sample.txt containing: "Data science is fun. Data is useful."
Word count: 7
```